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# Patterns of analogical reasoning among beginning readers

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Despite compelling evidence that analogy skills are available to beginning readers, few studies have actually explored the possibility of identifying individual differences in young children's analogy skills in early reading. The present study examined individual differences in children's use of orthographic and phonological relations between words as they learn to read. Specifically, the study addressed whether general analogical reasoning, short-term memory and domain-specific reading skills explain 5- to 6-year-olds' reading analogies ($n = 51$). The findings revealed an orthographic analogy effect accompanied by high levels of phonological priming. Single-word reading and use of visual analogies predicted young children's orthographic and phonological analogies in the regression analyses. However, different findings emerged from exploring profiles based on individual differences in reasoning skill. Indeed, when individual differences in composite scores of orthographic and phonological analogy were examined, group membership was predicted by word reading and early phonological knowledge, rather than general analogical reasoning skills. The findings highlight the usefulness of exploring individual differences in children's analogy development in the early stages of learning to read.

The aim of the study was to explore patterns of individual differences in children's use of orthographic and phonological analogies and to investigate the extent to which domain knowledge and general analogical reasoning skills contribute to these patterns of individual differences. Despite a growing recognition of the potential importance of orthographic analogies in early reading, there has been a tendency for research to concentrate on age-related differences and to treat beginning readers as an homogeneous group (e.g. Goswami, 1986, 1988, 1990; Goswami & Mead, 1992; Muter, Snowling & Taylor, 1994; Wood, 1999, 2000, 2002) rather than considering the potential scope for individual differences in how they cope with making analogies during reading. We recommend that exploring and comparing different profiles of analogical reasoning skills will provide a more detailed understanding of the mechanisms responsible for children's

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success on such tasks. Furthermore, although it has been hypothesised that children's developing levels of phonological awareness can predict patterns of individual differences in young children's orthographic analogies (Goswami, 1990, 1993), relatively little is known about whether other forms of analogical reasoning skill, such as their ability to recognise visual similarity or common themes, contribute to children's analogical success on the clue word task. These issues were addressed in the current study.

One of the most prominent accounts of orthographic analogy development is the interactive analogy model of early reading (Goswami, 1993). The premise of this model is that children's ability to use analogies in early reading is linked to the development of phonological awareness and shows that children's progression in learning to read develops along a path of refined use of lexical analogies. During the early stages of learning to read, when children are introduced to single syllable words, they tend to associate spelling sequences with two large phonological units (i.e. onset-rime units) and so the first type of analogy to emerge is one which represents these phonological structures. As children's phonological knowledge improves they develop an understanding of individual phonemes and graphemes and a more sophisticated level of phonemic representation, which allows them to make increasingly refined orthographic analogies. According to this account, the development of analogy in reading will therefore be affected by this refinement in phonological awareness (Goswami, 1993).

Many studies have provided empirical support for the interactive analogy model, most of which have relied on the clue word paradigm as a measure of analogical transfer in beginning reading (Goswami, 1988, 1990, 1993; Goswami & Mead, 1992; Muter, Snowling & Taylor, 1994; Nation, Allen & Hulme, 2001; Savage & Stuart, 2001; Walton, 1995; Wood, 2000). Within this task, children are presented with a clue word, told how to pronounce this word and are then shown a series of unfamiliar words, some of which share an orthographic overlap with the clue word (e.g. bean–mean). Children are then presented with a series of target words, presented one at a time, and asked to read these back to the experimenter correctly. The clue word remains visible throughout the procedure allowing children to refer back to its spelling-sound pattern. This procedure is built on the assumption that if children are sensitive to the orthographic overlap between clue and test words, then the mere presence of the clue word should be sufficient to provide a prompt that will allow the child to provide the correct pronunciation of a similarly spelled word.

Goswami and colleagues (Goswami, 1986, 1988, 1990; Goswami & Mead, 1992), in a series of early prominent investigations, found evidence of a reliable orthographic analogy effect in beginning reading, in that young children read significantly more orthographically related items compared to unrelated controls. Further analyses revealed that beginning readers were better at making orthographic rime analogies than beginning or medial-vowel analogies (Goswami, 1993). These findings led to the initial conclusion that children consciously attend to the orthographic overlap between clue words and test words and use this overlap as a basis for generating the correct pronunciation of unfamiliar words. In particular, children's success in using orthographic analogies in early reading is dependent on the child's developing phonological awareness of spoken language.

There is fairly widespread agreement that analogical reasoning skills in the context of early reading are constrained by children's understanding of those principles that govern that particular domain. In keeping with arguments that domain relevant knowledge underpins analogical success in early reading, research suggests that both children's exposure to print and receptive vocabulary (Bowey & Francis, 1991; Bowey &

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Underwood, 1996) and their appreciation of phonological spelling-sound units in early reading (Goswami, 1993; Goswami & Bryant, 1990; Walton, 1995; Wood, 1999, 2000), appear to play an influential role in governing children's success in using an orthographic analogy reasoning strategies in early reading. Furthermore, it is apparent that different types of phonological awareness, corresponding to aspects of children's awareness of large phonological units, such as onset-rime (Goswami & Bryant, 1990; Goswami & Mead, 1992) and smaller phonological units such as phonemes (Ehri & Robbins, 1992; Roberts & McDougall, 2003; Walton, 1995; Wood, 2000; Wood & Terrell, 1998) are strong simultaneous predictors of orthographic analogies in beginning reading. Taken together, these data provide convergent evidence that analogical reasoning in the context of reading is, in part, constrained by children's knowledge of phonological structures within that particular domain.

There are, however, concerns over the validity and reliability of the orthographic analogy effect in early reading. Some researchers question whether rime is the most important aspect of the orthographic prompt compared to other parts of the clue word (Bowey, Vaughan & Hansen, 1998; Brown & Deavers, 1999; Bruck & Treiman, 1992; Nation, Allen & Hulme, 2001). Other researchers have suggested that children's ability to make orthographic analogies in reading is dependent on task design and that the use of analogy cannot be generalised beyond the confines of the task (e.g. Brown & Deavers, 1999; Muter, Snowling & Taylor, 1994; Savage, 2001). Furthermore, it has been argued that hearing the phonological pronunciation of the clue word alone is sufficient to facilitate analogical transfer. According to findings from a series of recent studies, the apparent orthographic analogy effects in early reading may be best explained by phonological priming (Bowey, Vaughan & Hansen, 1998; Nation, Allen & Hulme, 2001; Roberts & McDougall, 2003; Wood, 2002). The difficulty lies in attempting to establish whether there is any validity to the phonological priming explanation (see Bowey, 1999; Goswami, 1999b).

There has been a significant number of research studies designed to examine the role of phonological priming in orthographic analogy effects in beginning reading. In a key study, Bowey, Vaughan & Hansen (1998) compared beginning readers' performance on orthographic rime items (bone–cone) and phonological prime items (cone–moan). Their findings suggest that children were not using a genuine orthographic strategy to read analogous words, but relied on a rhyme-based phonological prime strategy. Further evidence for the phonological priming explanation comes from recent studies designed to manipulate the nature of the prompt and the target words in the clue word task (Nation, Allen & Hulme, 2001; Savage & Stuart, 1998, 2001). These studies reveal equivalent levels of performance regardless of whether children are given an orthographic or a phonological prompt (Nation, Allen & Hulme, 2001). However, the idea that the children's responsiveness to phonological priming (in the context of the clue word task) might be of interest in its own right, as indicative of a level of developing phonological competence, has not yet been fully examined. The importance of examining this aspect of children's performance becomes all the more significant if we change our approach to analysing the results of such studies to enable a discussion of individual differences in children's development of an orthographic analogy strategy.

Despite the prominence of the interactive analogy model of early reading (Goswami, 1993), few studies have addressed the possibility of individual differences in children's analogical skills. At least theoretically, there are suggestions that individual differences in reading-based analogical skills may exist (see Bowey, Vaughan & Hansen, 1998;

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Goswami, 1988, 1990, 1993). In a series of early investigations, Goswami (1988, 1990) showed that the development of analogies in word reading was associated with developing phonological knowledge of sounds. The suggestion is put forward that individual differences in children's analogies are related to varying degrees of early phonological awareness (see also, Goswami, 1993). However, we argue that what these early studies actually show is a linear trend in analogical reasoning skills developing in line with age. The claims regarding individual differences are based solely on age-related differences: five-year-olds are accurate in making orthographic rime analogies, six-year-olds are accurate in making beginning analogies and analogies based on shared vowel diagraphs (Goswami, 1993). There are no analyses of variation *within* these specific age groups because only mean scores between different age groups are examined. Concentrating solely on group-level rather than individual-level performance is problematic because this ignores considerable differences among children of the same age and leads to inaccurate or at least incomplete descriptions of children's reading abilities (see also Siegler, 1988). Given that the theoretical approach is not set up to explore individual differences in reasoning skills explicitly, we really do not fully understand what is happening when children solve analogical problems in beginning reading. Instead there appears to be a tendency to make generalised claims about children's abilities and performance because analyses are based on age-related changes in reasoning ability alone. However, development does not always proceed in a linear, age-related fashion, and it is entirely possible that some children may find their own routes to reading competence based on their general cognitive abilities and their level of domain-relevant knowledge. Moreover, in other areas of development there are critical points where there is an unexpected qualitative shift in how children approach a given task which often results in children exhibiting inappropriate strategies, thereby demonstrating age-related, but non-linear, patterns of development. However, at the moment these types of detailed analysis are absent from accounts of orthographic analogy use.

The question remains whether studying patterns of variation in children's reasoning abilities will offer any further useful contribution to existing theories of reading development (e.g. Goswami, 1993; Goswami & Bryant, 1990). We recommend that studying individual differences in children's analogy will provide a detailed and more comprehensive account of the ways in which children use analogical reasoning skills in early reading. Using analytical methods that explore different profiles of reasoning skills allows us to explore the phonological underpinnings for different groups of children and this in turn may signal different developmental pathways to reading success. An expectation of the present study was that different patterns of reasoning skills will emerge and that these will vary according to children's phonological knowledge (Goswami, 1993, 1998).

The extent to which more general forms of analogical reasoning ability might contribute to the ability to make orthographic analogies in reading has also not been explored systematically. It is currently unclear whether other forms of analogical reasoning ability can account for children's use of orthographic analogies and/or susceptibility to phonological priming during reading. This is an important question because it addresses the question of whether analogy use in early reading is determined solely by a child's level of domain knowledge or whether it is determined in part by a more general understanding of analogy. Relations between general analogical reasoning ability and orthographic analogy use in beginning readers have been reported in one study. Wood (1999) explored whether measures of single word reading ability, phonemic awareness and visual analogical reasoning ability could predict the success beginning

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readers have in making orthographic analogies. Stepwise regression analyses found that phoneme awareness was able to account for 40% of the variance in children's orthographic analogy use, and children's performance on the visual analogical reasoning task could contribute an additional 5% to the overall variance. Given the nature of Wood's (1999) findings, it is reasonable to suggest that other forms of traditional reasoning tasks may predict the use of orthographic analogies in early reading. This claim awaits empirical support.

An interesting extension to Wood's (1999) research is to explore whether other forms of general analogical reasoning skills are related to individual differences in children's orthographic and phonological analogies. In particular, it is not known whether differences in such abilities might be able to discriminate between children who are successful and those who are less successful in making orthographic analogies. It is likely that beginning readers vary in their ability to use orthographic analogies in early reading (with respect to both speed and accuracy of performance) and in their levels of phonological awareness. Moreover, the extent to which profiles of individual differences in orthographic analogy use and phonological priming are related to more general analogical reasoning skills is unknown and requires further investigation. It may be the case that children who are more proficient at solving orthographic analogies may also show more advanced levels of general analogical reasoning ability compared with children who are less able to make orthographic analogies or who demonstrate a greater susceptibility to phonological priming.

The present study was designed to address four specific questions. First, is orthographic analogy of salience to beginning readers and can it be distinguished from phonological priming effects? Second, to what extent is children's proficiency in making orthographic analogies in early reading related to aspects of more general forms of thematic, causal and visual reasoning ability? Third, are there meaningful patterns of individual differences in the speed and accuracy with which young children can make analogies? Fourth, to what extent do children's general analogical reasoning skills, reading ability and phonological knowledge explain profiles of performance on the clue word task?

## Method

### *Participants*

Fifty-one children, with an age range of 5 years 1 month to 6 years 4 months (Mean = 5 years 6 months, *SD* = 4 months), participated in the study. The children were selected from two schools in Nottinghamshire and were tested towards the beginning of the academic year. The mean reading ability score of the children on the British Ability Scales single word reading subtest was 94.08 (*SD* = 7.11), which equates to an approximate reading age of 5 years 8 months. The mean score of the group on the British Picture Vocabulary Scales was within normal limits (103.84, *SD* = 9.39). All children spoke English as their first language. Full informed parental consent was obtained for the children's participation in the study.

### *Initial pre-test and screening*

All children who were in Reception and Year 1 classes ($n = 138$) were screened and recruited according to their level of reading proficiency. Single word reading ability was assessed using the British Ability Scales single word reading subtest in order to establish

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that children could demonstrate some emergent reading ability. Children were then pre-tested on their ability to read the words that would be used in the clue word task. To be included in the study, children had to demonstrate initial reading ability, but only those children who were unable to read any of the words that were to be used in the clue word task were selected ($n = 51$). This ensured that all children began the study within the same limits of competence in reading and prevented the need to calculate pre-test and post-test analogy scores (see Wood, 1999, 2000, 2002).

*Receptive vocabulary*

The British Picture Vocabulary Scales is similar in design to the Peabody Picture Vocabulary Scales and is a measure of receptive vocabulary, which requires children to identify pictures that correspond to words spoken by the experimenter. The words become progressively more difficult and children are required to continue until they can only get four words correct out of a set of twelve items. The raw score is then converted to a standardised score (Mean = 100, $SD = 15$) for the purpose of analysis (Dunn, Dunn, Whetton & Burley, 1997).

*Backward digit-span*

The backward digit-span taken from the British Ability Scales II was included in the study as a measure of short-term memory. This task consists of six trials, each trial containing five items in total. Each trial increases in difficulty. The children are given a sequence of digits and are required to repeat that sequence backwards. Digits are presented at a regular pace with a two-second interval between the presentations of each digit. Children are given one practice trial where feedback is provided in order to familiarise them with the task. One point is scored for each sequence repeated correctly (Elliot, Smith & McUlloch, 1996).

*Rhyme Awareness Task*

The Rhyme Awareness Task is included in the study to provide a measure of children's sensitivity to rhyme. In this task the children hear three words spoken, two of which rhyme. The children are asked to say which two words rhyme. They complete four practice items (during which feedback is provided) followed by up to 21 test items that increase in difficulty. The children score one point for each pair of words named correctly (Frederickson, Frith & Reason, 1997).

*Phonemic awareness*

The children's understanding of phoneme units was assessed using a phoneme deletion task, which requires the child to delete either the initial or final phoneme from a word to create a new word: e.g. 'Try to say ''car'' without saying the /k/'. The task begins with three practice items for the initial phoneme deletion sub-scale, followed by the actual test items. Corrective feedback is allowed during the practice items to ensure the child understands the instructions, but no feedback is given during the actual test items. The same sequence is followed for the final deletion sub-scale. There are twelve items in the initial and final phoneme sub-scale, respectively. The children receive one point for each correct deletion made (Wood, 1999).

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*General analogical reasoning tasks*

A series of traditional analogical reasoning tasks are included in the study to provide a measure of children's general analogy skill. This comprised of three separate reasoning tasks designed to assess children's ability to complete pictorial A:B::C:? analogies based on familiar relations. Each of the three tasks used the same experimental procedure but varied in terms of the types of relations that children were required to use in order to complete problems successfully (see Figures 1, 2 and 3). These tasks were presented on a computer laptop which allowed sensitive measures of accuracy and speed to be recorded. (The speed and accuracy scores were adjusted to take into account the different number of choices available on each of the three analogical reasoning tasks. These scores were weighted by multiplying the scores obtained by the total number of choices available.)

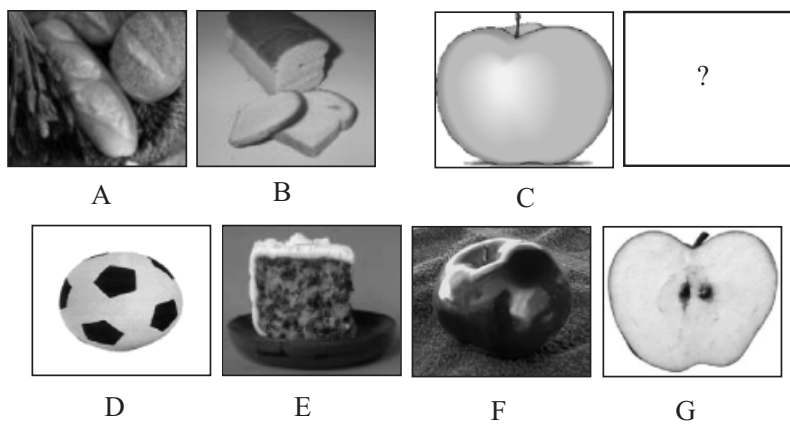


**Figure 1.** An example of the stimuli used in the causal reasoning task (*bread is to slice of bread as apple is to slice of apple*) including the correct answer and possible distractors.

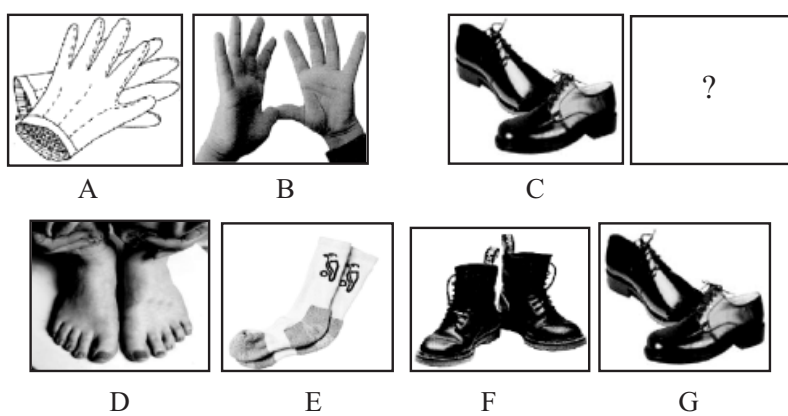


**Figure 2.** An example of the stimuli used in the thematic reasoning task (*gloves are to hands as shoes are to feet*) including the correct answer and possible distractors.

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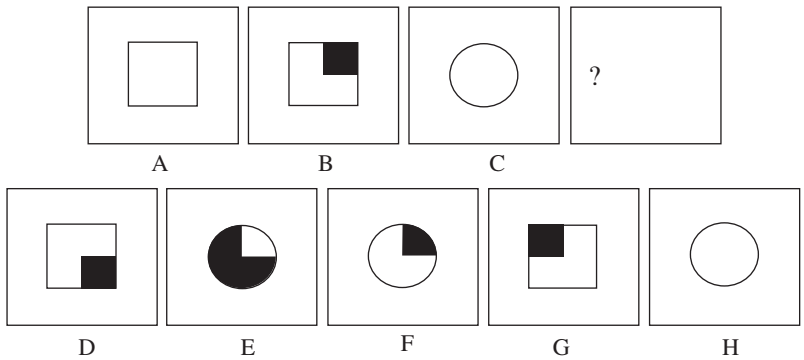


**Figure 3.** An example of the stimuli used in the visual reasoning task (*square: 1/4 square shaded: circle: 1/4 circle shaded*) including the correct answer and possible distractors.

*The causal reasoning task*. The task assessed young children's ability to use knowledge of physical causal relations (such as 'melting' or 'cutting'), as a basis for completing analogies in the form of *chocolate is to melted chocolate as snowman is to puddle* (Goswami, 1989a). This task comprises ten trials in total. Children are presented with a sequence of three pictures (e.g. *chocolate*, *melted chocolate* and *snowman*) and are asked to choose a fourth picture to complete the sequence from a range of four alternative choices (e.g. the correct answer and perceptual, category and associative distractors). One point is scored for each sequence solved correctly (out of a possible ten).

*The visual reasoning task*. This task assessed children's ability to complete analogies on the basis of shared visual relations (Wood, 1999). This task comprises ten trials in total. Children are asked to look closely at the different shapes, compare them and explain how the shaded part had changed. Children are presented with a sequence of three pictures and asked to choose a fourth picture to complete the sequence from five alternative choices (e.g. correct answer and four distractors). One point is scored for each sequence solved correctly (out of a possible ten).

*The thematic reasoning task*. The task assesses young children's ability to use their knowledge of thematic relations (such as 'lives in') to complete analogies in the form of *dog is to kennel as bird is to nest* (Goswami & Brown, 1990). This task consists of ten trials in total. Children are presented with a sequence of three pictures (e.g. *dog*, *kennel*, *bird*) and asked to choose a fourth to complete the sequence from four alternative choices (e.g. correct answer and three distractors: wrong object-correct transformation; correct object-wrong transformation and a high similarity object match). One point is scored for each sequence solved correctly (out of a possible ten).

### *The Reading Analogy Task*

The Reading Analogy Task is a revised version of the traditional clue word task (e.g. Goswami, 1986, 1988, 1990). The task is included in the study to provide a measure of domain-specific analogy skill in beginning reading. The task assesses children's ability to

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use the spelling-sound pattern of one word as a basis for working out the pronunciation of the similarly spelled words. A computer version of the task was devised in order to obtain a measure of the accuracy and speed of each individual response. During the task, the clue word appeared in the centre of the screen and the child was told its pronunciation and told that this clue word might help them to read other words. The experimenter hit a timing key to present a new target word and children were asked if they were able to read this particular target word. The original clue word remained in view, which enabled the child to refer back to the spelling pattern of that particular word. The assessor hit a key when the child stated their answer in order to record solution times. (Pilot testing revealed that although a voice-activated timing system was the most accurate way to record solution times, these were unreliable given that children often spoke out loud while working out their answer).

Two sets of matched word lists were devised containing nonword items. The presentation of these two sets of word lists was given over two separate experimental sessions. This ensured that word sets were not repeatedly using the same vowel phonemes (see Bowey, Vaughan & Hansen, 1998; Goswami, 1999b). The inclusion of nonword items assisted in controlling for the effects associated with high word frequency and unintentional intralist priming found in previous research (see Bowey, Vaughan & Hansen, 1998; Wood, 2002; Wood & Farrington-Flint, 2002). Lastly, the conditions were also devised in a way that allowed us to explore the possibility that children might respond to the phonology of the clue word rather than the words' orthographic similarity. This was achieved by including items that were phonologically similar but orthographically dissimilar to the clue word items. The clue word task comprised twenty-one clue words and five different types of target words.

(i) *The orthographic rime analogy.* These words share a common phonological rime and orthographic rime unit with the clue word (*kurp–nurp*).
(ii) *The phonological rhyme analogy.* These words share a common phonological rime with the clue word, but a different orthographic pattern (*kurp–herp*).
(iii) *The orthographic beginning analogy.* These words share a common phonological and orthographic beginning as the clue word up to but not including the final phoneme (*kurp–kurn*).
(iv) *The phonological beginning analogy.* These words share a common phonological beginning up to, but not including, the last phoneme and similarly share a different orthographic pattern to the clue word (*kurp–kerf*).
(v) *Unrelated controls.* These words share no common phonological or orthographic overlap with the clue word and therefore using an analogy strategy would result in an incorrect response (*kurp–helt*).

## General procedure

Each child was seen individually on three separate occasions. In the pre-test session, following the British Ability Scales II single word reading test, the children were asked to read the list of words that were to be used in the reading analogy task. On the basis of being unable to read any of these pre-test words, the children were recruited to the study. The children then completed the baseline measures consisting of the British Picture Vocabulary Scales, Rhyme Awareness and Phoneme Deletion tasks in the first 30-minute session. They completed List A items in the Reading Analogy Task and the visual

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reasoning task in the second 30-minute session. In the final session, the causal and thematic analogical reasoning tasks and List B of the Reading Analogy Task was administered. The presentation of tasks was counterbalanced across participants.

## Results

The results are presented in four parts. In the first part, the salience of orthographic analogies to young beginning readers was explored. The extent to which the children were able to make orthographic analogies compared to their performance on the phonological prime items was investigated. In the second part, we examined whether children's single word reading, phonological knowledge and their general analogical reasoning scores could predict their use of orthographic and phonological analogies in early reading. In the third part, individual differences in young children's use of analogies in beginning reading were explored. Specifically, the possibility that individual differences underlie children's ability to solve orthographic (and phonological) analogies in beginning reading was examined using cluster analysis. In the fourth part, the extent to which reading ability, phonological knowledge and general analogical reasoning skill, could explain individual differences in domain-specific analogy, was examined.

### *Orthographic analogies and phonological priming*

The goal of the first set of analyses was to examine whether orthographic analogy is of salience to young beginning readers and whether this aspect of early reasoning skill can be distinguished from the effects associated with phonological priming. Two separate analyses were conducted to investigate the nature of analogical scores according to the accuracy and speed of children's responses across the five different conditions. Analyses were conducted across subjects and across items. The mean scores and standard deviations for the speed and accuracy of children's performance across the five conditions are summarised in Table 1.

The data were analysed using a one-way within-groups analysis of variance. In the accuracy analyses, the results showed a main effect for analogy type, by-subjects: $F(4, 50) = 77.81$, $p < 0.001$, $\eta^2 = 0.607$; by-items: $F(4, 75) = 59.98$, $p < 0.001$, $\eta^2 = 0.762$. Planned comparisons revealed that orthographic rime words were read more frequently than the phonological rime words, by-subjects: $t(50) = 3.18$, $p < 0.001$, although by-items, this effect was not found to be significant, by-items: $t(30) = 2.01$, $p = 0.053$. As expected, children were more proficient at reading orthographic rime words than the orthographic beginning words, by-subjects: $t(50) = 4.08$, $p < 0.001$; by-items: $t(30) = 3.74$, $p < 0.001$. They were also more proficient at reading orthographic beginning words than the phonological equivalents, by-subjects: $t(50) = 5.17$, $p < 0.001$; by-items: $t(30) = 3.01$, $p < 0.01$.

To explore whether different patterns of analogical performance emerge on the basis of children's individual solution times, a second analysis of variance was conducted. (Analyses only included individual solution times of less than 30 seconds and excluded incorrect responses.) The solution time analyses showed a main effect for analogy type by-subjects, $F(4, 50) = 2.65$, $p < 0.01$, $\eta^2 = 0.050$, although not by-items, $F(4, 75) = 2.00$, $p > 0.05$, $\eta^2 = 0.010$. Planned comparisons by-subjects showed that the children were no quicker at reading words sharing orthographic rimes than those with

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**Table 1.** Means (and standard deviations) for the speed and accuracy of children's scores on the Reading Analogy Task and the General Analogical Reasoning Tasks.

| | Accuracy | Speed | α |
|---|---|---|---|
| The Reading Analogy Task (max. 16) | | | |
| Orthographic rime analogy | 10.37 (3.28) | 5.67 (3.50) | 0.71 |
| Phonological rhyme analogy | 9.31 (3.67) | 5.68 (3.65) | 0.77 |
| Orthographic beginning analogy | 8.12 (3.81) | 7.04 (4.25) | 0.79 |
| Phonological beginning analogy | 6.04 (3.94) | 7.28 (5.78) | 0.82 |
| Unrelated controls | 2.33 (2.58) | 5.43 (5.32) | 0.75 |
| General Analogy tasks (max. 10) | | | |
| Thematic reasoning | 5.51 (1.45) | 4.82 (1.65) | 0.61 |
| Causal reasoning | 5.90 (1.53) | 4.38 (1.51) | 0.52 |
| Visual reasoning | 4.10 (1.72) | 8.04 (3.14) | 0.55 |

*Note*: Solution-time in seconds (analyses include individual solution times of less than 30 seconds and excluded incorrect responses).

purely phonological rimes, $t(50) = -0.03$, $p > 0.05$ and no quicker at reading the orthographic beginning words than those words with purely phonological beginnings, $t(50) = -0.37$, $p > 0.05$. However, the children did respond quicker in reading the orthographic rime words in comparison to the orthographic beginning words, $t(50) = -2.42$, $p < 0.05$.

In order to examine the interrelations between orthography and phonology further, composite measures of orthographic and phonological reasoning skills were calculated by averaging the *z*-scores for relevant measures. Scores for the *orthographic rime* words and *orthographic beginning* words were combined to provide a composite measure of *orthographic analogy reasoning* skill. Scores for the *phonological rhyme* words and the *phonological beginning* words were combined to provide a composite measure of *phonological analogy reasoning* skill. Cronbach's alpha internal reliability coefficients for the new composite measures were at an acceptable level (composite orthographic reasoning, 0.81; composite phonological reasoning, 0.86). All subsequent analyses were performed using these composite scores.

### *Concurrent prediction of domain-specific analogy*

The goal of the second set of analyses was to address covariations between children's domain-specific analogy skills, general analogical reasoning skills and their phonological knowledge. In particular, we were interested in assessing the power of general analogical reasoning scores and phonological knowledge as concurrent predictors of orthographic and phonological analogy skills in early reading.

Before examining concurrent predictors of orthographic and phonological analogies, we examined raw correlations between single word reading ability, receptive vocabulary, short-term memory, general analogical reasoning skill and orthographic and phonological analogies. The correlation matrix, reported in Table 2, outlines the relationship between each of the individual variables. As expected, there are strong significant correlations between single word reading, rhyme awareness and phoneme awareness and each of these measures is strongly related to children's orthographic analogy scores. Also, it is interesting to note that measures of single word reading and phonological knowledge are all significantly correlated with our composite measure of phonological reasoning score,

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**Table 2.** Correlations between domain-specific analogical skills, phonological skills and general analogical reasoning skills.

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| 1. Orthographic analogies | – | | | | | | | | | |
| 2. Phonological analogies | 0.820** | – | | | | | | | | |
| 3. Single word reading | 0.531** | 0.501** | – | | | | | | | |
| 4. Vocabulary | 0.170 | 0.198 | 0.361** | – | | | | | | |
| 5. Short-term memory | 0.080 | −0.062 | 0.232 | 0.273 | – | | | | | |
| 6. Rhyme awareness | 0.449** | 0.465** | 0.541** | 0.328* | 0.122 | – | | | | |
| 7. Phoneme awareness | 0.534** | 0.583** | 0.499** | 0.153 | 0.029 | 0.527** | – | | | |
| 8. Thematic reasoning | 0.316* | 0.261 | 0.056 | −0.040 | 0.053 | 0.350* | 0.228 | – | | |
| 9. Causal reasoning | 0.227 | 0.331* | 0.205 | 0.051 | −0.261 | 0.273 | 0.401* | 0.041 | – | |
| 10. Visual reasoning | 0.389** | 0.433** | 0.016 | −0.031 | −0.162 | 0.103 | 0.183 | 0.060 | −0.042 | – |

*Note*: $^{*}p<0.05$; $^{**}p<0.001$.

although sadly correlations for short-term memory and receptive vocabulary scores were particularly low and neither of these measures could significantly predict either measure of orthographic or phonological forms of reasoning by analogy. However, there are strong systematic correlations between general analogical reasoning measures (causal, thematic and visual reasoning) and children's orthographic and phonological reasoning scores, which suggest that aspects of general analogical reasoning skills may be strong predictors of domain-specific analogising in the context of reading. The contribution of phonological factors and general analogical reasoning skills were examined further using hierarchical regression analysis.

To assess the power of general analogical reasoning ability and early phonological knowledge as predictors of domain-specific analogising, two series of hierarchical regression analyses were carried out with composite orthographic and phonological analogy scores as predictors. The order in which the general analogical reasoning tasks were entered into the regression model was systematically varied to allow us to examine the extent to which certain forms of analogical reasoning tasks act as unique predictors of reading analogy when the effects of other domain knowledge measures had been systematically controlled. (Given the low ratio between cases and independent variables, the regression analyses may show reduced power and the results should be interpreted with caution.)

In the first analysis, the composite orthographic analogy scores were treated as the dependent variable with age, single word reading, phonological scores and general analogical reasoning scores as predictors. Age was entered at step 1 and single word reading, rhyme awareness and phoneme awareness at steps 2 to 4. The causal, thematic and visual reasoning scores were entered into the regression model in all possible combinations at steps 5 to 7. (BPVS vocabulary and short-term memory scores were not entered into the regression model because they failed to account for any appreciable variance in orthographic and phonological analogy scores.) The results from this analysis are provided in Table 3 and show that age can account for a significant amount of variance when entered into the model at step 1 (14%, $p<0.01$). As expected, measures of single word reading (21%, $p<0.01$) can account for significant variance in orthographic analogy skills after the contribution of age has been systematically controlled. The only measure of analogical reasoning skill that can account for significant unique variance was visual reasoning skill (5%, $p<0.05$).

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**Table 3.** Hierarchical regression analyses with composite orthographic and phonological analogy scores as the dependent variable and age, domain knowledge and general analogical reasoning scores as predictors.

| Steps 1 to 7 | Orthographic reasoning | | | Phonological reasoning | | |
|---|---|---|---|---|---|---|
| | $R^2$ | $R^2$ change | $F$ value | $R^2$ | $R^2$ change | $F$ value |
| 1. Age (months) | 0.14 | 0.14 | $8.04^{**}$ | 0.21 | 0.21 | $12.96^{**}$ |
| 2. Word reading | 0.37 | 0.23 | $17.54^{**}$ | 0.42 | 0.21 | $17.63^{**}$ |
| 3. Rhyme skills | 0.40 | 0.03 | 2.17 | 0.45 | 0.03 | 2.83 |
| 4. Phoneme skills | 0.44 | 0.04 | 3.17 | 0.51 | 0.06 | $5.14^{*}$ |
| 5. Thematic reasoning | 0.46 | 0.02 | 1.95 | 0.51 | 0.00 | 0.36 |
| 6. Causal reasoning | 0.46 | 0.00 | 0.06 | 0.52 | 0.00 | 0.20 |
| 7. Visual reasoning | 0.42 | 0.06 | $5.39^{*}$ | 0.59 | 0.07 | $7.47^{**}$ |

*Note*: $^{*}p<0.05$; $^{**}p<0.001$.

In the second series, phonological reasoning scores were treated as the dependent variable and phonological knowledge and general analogical reasoning scores as predictors. A similar pattern of findings emerged to the previous analysis. Age accounted for a significant amount of variance (21%, $p<0.01$), followed by single word reading (21%, $p<0.01$). Phoneme awareness was also found to account for an additional 6% after age and single word reading had been entered into the regression model. The only unique predictor of phonological analogy score in reading, after all other measures had been taken into account was visual reasoning skills (7%, $p<0.01$).

### *Individual differences in domain-specific analogy*

The previous analyses examined the relationship between orthographic and phonological analogy skills by comparing aggregated data across the sample without fully considering the possibility of individual differences in scores. Therefore, the goal of this set of analyses was to explore whether meaningful patterns of individual differences in the speed and accuracy with which young children perform analogies could be identified. Cluster analysis was used because this is considered a useful way of identifying characteristic patterns among different children and often provides a detailed picture of children's competencies across tasks (Siegler, 1988). Distinct patterns of analogical skills were identified using Wards clustering algorithm. The clustering algorithm was applied to the number of analogy words read correctly and the average solution times on the clue word task (using the composite orthographic and phonological scores). Standardised scores were used in order to take into account different weightings of variables and to ensure an equal contribution from the accuracy and solution time data. The results from the cluster analysis are summarised in Table 4 and show that the clusters are ordered. (Discriminant function analysis revealed that the groups were significantly discriminated by all predictors.)

A four-cluster solution, which accounted for 75% of the total variation in children's scores, was selected. Children in Cluster 1, named the *high reasoning group* ($n = 14$), were adept at recognising both orthographic and phonological analogies, responding both efficiently and quickly to both types of analogies in the clue word task. Equivalent levels of transfer across both conditions suggests that children were using a phonological rhyming strategy to read the analogous words, rather than making connections between shared spelling patterns and shared sound. Children in Cluster 2, named the *intermediate*

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**Table 4.** Means (and standard deviations) of composite orthographic and phonological analogy scores as a function of cluster.

| | Domain-specific analogy cluster | | | |
|---|---|---|---|---|
| | High reasoning | Intermediate reasoning | Orthographic reasoning | Low reasoning |
| Accuracy scores | | | | |
| Orthographic analogies | 26.14 (2.21) | 17.19 (2.14) | 17.50 (4.33) | 11.55 (2.51) |
| Phonological analogies | 23.14 (3.37) | 16.13 (3.86) | 11.90 (4.51) | 7.45 (2.58) |
| Solution time (in seconds) | | | | |
| Orthographic analogies | 3.57 (0.55) | 4.35 (1.12) | 9.75 (2.50) | 9.74 (2.47) |
| Phonological analogies | 3.75 (0.64) | 4.78 (1.29) | 12.20 (3.58) | 7.23 (2.72) |
| *n* | 14 | 16 | 10 | 11 |

*Note*: Solution-time analyses of less than 30 seconds and excluded incorrect responses.

*reasoning group* ($n = 16$), showed a similar pattern of results to the *high reasoning group* although they were less accurate and slower in reading the analogous words and phonological primes than children in the preceding cluster. By comparison, in Cluster 3, named the *orthographic reasoning group* ($n = 10$), children were generally slower in both the orthographic and phonological conditions, but in terms of orthographic accuracy, there is little difference between the *intermediate reasoning* and *orthographic reasoning* groups. Children were less able to read words in the phonological condition. This advantage for reading orthographic analogy words provides an indication that the children in this group were reading the analogous words on the shared orthographic overlap between words (orthographic strategy) rather than simply applying a phonological rhyming strategy across all conditions (phonological prime). Children in Cluster 4, named the *low reasoning group* ($n = 11$), were slower in the orthographic and phonology analogy conditions although they were more accurate in reading analogous words than phonological primes. Their low levels of accuracy suggest that these children may have been using inaccurate or illegitimate strategies that often led them to the incorrect pronunciations of words.

Furthermore, individual differences in children's analogy scores were not due to age-related changes in the sample, $F(3, 47) = 2.10$, $p > 0.05$.

### *Relations between children's domain-specific analogy, general analogical reasoning skills and their domain knowledge*

The goal of the fourth set of analyses was to examine relations between profiles of individual differences in children's domain-specific analogy, general analogical reasoning skills and their phonological knowledge. To establish the extent to which children's knowledge of the reading domain and their general analogical reasoning skills can explain patterns of individual differences in domain-specific analogising in reading; two direct discriminant function analyses were conducted (see Tables 5 and 6).

The first discriminant function examined relations between measures of this analogical component of reading and other forms of general analogical reasoning task. We examined whether children's performance on the thematic, causal or visual analogy reasoning tasks could explain our pattern of results in children's reading analogy. Predictors were

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**Table 5.** Means (and standard deviations) for general analogical reasoning scores as a function of domain-specific cluster.

| General reasoning tasks | Domain-specific analogy cluster | | | |
|---|---|---|---|---|
| | High reasoning | Intermediate reasoning | Orthographic reasoning | Low reasoning |
| Thematic reasoning | 24.29 (5.31) | 21.75 (5.84) | 22.00 (5.73) | 19.64 (6.05) |
| Causal reasoning | 25.14 (6.16) | 23.75 (6.45) | 24.40 (4.79) | 20.73 (6.41) |
| Visual reasoning | 23.21 (7.99) | 21.87 (10.31) | 19.00 (8.10) | 16.36 (5.94) |
| *n* | 14 | 16 | 10 | 11 |

**Table 6.** Means (and standard deviations) of single word reading and phonological knowledge as a function of domain-specific analogy cluster.

| Domain knowledge | Domain-specific analogy cluster | | | |
|---|---|---|---|---|
| | High reasoning | Intermediate reasoning | Orthographic reasoning | Low reasoning |
| BAS II single word reading | 9.71 (4.79) | 7.13 (4.69) | 4.10 (1.79) | 2.73 (1.27) |
| Rhyme awareness | 12.79 (2.19) | 10.50 (4.23) | 8.80 (4.32) | 6.91 (2.95) |
| Phoneme awareness | 14.64 (2.76) | 11.81 (4.45) | 10.90 (4.58) | 8.27 (5.33) |
| *n* | 14 | 16 | 10 | 11 |

frequency of correct responses on the thematic relations, causal relations and visual relations task. Groups consisted of high reasoning, intermediate reasoning, orthographic reasoning and low reasoning, as indexed by cluster membership. A total of 51 cases were analysed. As Table 5 shows, univariate ANOVAs revealed that none of the groups differed significantly on each of the three predictors: for thematic relations, $F(3, 47) = 1.38$, $p > 0.05$, for causal relations, $F(3, 47) = 1.18$, $p > 0.80$, or for visual relations, $F(3, 47) = 1.60$, $p > 0.05$. Three discriminant functions were calculated. Analyses revealed that the combined function, $\chi^2(9) = 12.24$, $p > 0.05$ was significant. After removal of the first function, $\chi^2(4) = 0.90$, $p > 0.05$ and removal of the second function, $\chi^2(1) = 0.27$, $p > 0.05$, there was no significant association between groups and predictors. Clearly, individual differences in children's orthographic and phonological reasoning skills in early reading could not be explained in terms of their general analogical reasoning ability.

A second discriminant function analysis examined the extent to which children's knowledge of the reading domain, specifically, their single word reading and phonological knowledge, could explain patterns of individual differences in reading analogy. (Short-term memory and BPVS vocabulary scores were not included in the discriminant function because of their poor correlation with the outcome measures.) Predictors were children's scores on the BAS single word reading, rhyme awareness and phoneme awareness tasks. The results are summarised in Table 6. Univariate ANOVAs revealed that each group differed significantly on each of the three predictor variables: single word reading, $F(3, 47) = 8.44$, $p < 0.01$, rhyme awareness, $F(3, 47) = 6.22$, $p < 0.01$, and phoneme awareness, $F(3, 47) = 4.63$, $p < 0.01$. On the basis of these

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findings, three discriminant functions were calculated. With all three functions included the model revealed significant relationship between groups, $\chi^2(9) = 26.57$, $p < 0.01$, and predictors which accounted for 97.1% of the overall between group variability. After removal of the first discriminant function, $\chi^2(4) = 1.01$, $p > 0.05$, and removal of the second function, $\chi^2(1) = 0.01$, $p > 0.05$, this relationship was no longer reliable. The first discriminant function maximally separates the orthographic reasoning group from the three remaining clusters (the high reasoning, intermediate reasoning and low reasoning groups) and correlations between predictors and discriminant function suggest that measures of single word reading ($r = 0.86$), rhyme awareness ($r = 0.74$) and phoneme awareness ($r = 0.62$) were all reliable predictors of group membership. The discriminant function successfully predicted 54.9% of cases with accurate predictions being made for 70.0% of scores in the orthographic reasoning group, 64.3% of scores in the intermediate reasoning group, 25.0% of scores in the high reasoning group and 72.7% of the low reasoning group. Overall, the results provide convergent evidence for a systematic relationship between domain-specific analogy and children's early phonological knowledge (Goswami, 1993).

## Discussion

The aim of the study was to explore individual differences in children's orthographic analogy use and susceptibility to phonological priming and to investigate the extent to which domain knowledge and general analogical reasoning skills contribute to these patterns of individual differences. Four main outcomes were identified. First, our findings demonstrate that young children have an analogy mechanism available to them in the early stages of learning to read. Although there was some evidence that children were able to make analogies between words that shared similar spelling-sound patterns, there was also evidence of phonological priming. Second, children's single word reading and visual analogy skills are unique concurrent predictors of orthographic and phonological analogies in reading. Third, distinct patterns of individual differences in children's use of orthographic and phonological analogy skills were identified using cluster analysis. Fourth, the only significant predictor of individual differences in beginning readers' use of analogies in word reading was their developing phonological knowledge.

The first analysis examined the salience of the orthographic analogy effect in beginning reading by comparing the accuracy and speed of reading orthographic versus phonological words. Although there was clear evidence that the children were making orthographic analogies between the beginnings of words ($p < 0.01$), their performance on the rime-related items in the task was more problematic. In three of the possible four planned comparisons, (by items/subjects, accuracy/speed), there were no significant differences in performance between orthographically similar word pairs and the phonologically similar word pairs. These findings imply that the children were concentrating solely on the pronunciation of clue word and were not attending to the orthographic overlap between clue and test words (see Bowey, Vaughan & Hansen, 1998; Nation, Allen & Hulme, 2001; Roberts & McDougall, 2003; Savage & Stuart, 1998, 2001 for discussion). Therefore, our finding that beginning readers were using phonological sounds rather than shared orthographic spelling sequence is incompatible with earlier interactive analogy models of reading development (Goswami, 1993). However, high

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levels of phonological priming do not show that young children are completely insensitive to the orthographic relationship between clue and target words. The availability of phonological primes in combination with varying degrees of orthographic knowledge may be jointly responsible for our results (see also Nation, Allen & Hulme, 2001). Children may be using a combination of orthographic and phonological correspondences to read target items in and the findings may simply demonstrate that some children analogise orthographically whilst others use phonological priming. The present study also examined concurrent predictors of domain-specific analogy skills in beginning reading. Previous research has showed how aspects of children's rhyme awareness, phoneme awareness and vocabulary skill are all important predictors of orthographic forms of analogical reasoning in beginning reading (Goswami, 1988, 1990, 1993; Goswami & Mead, 1992; Muter, Snowling & Taylor, 1994; Roberts & McDougall, 2003; Wood, 1999, 2000, 2002; Wood & Farrington-Flint, 2002). However, few research studies have explored the predictors of phonological priming effects in early reading. The results from the regression analyses showed that single word reading was an excellent predictor, accounting for 23% in orthographic analogy scores and 21% in children's phonological analogy scores in beginning reading. Surprisingly, children's phonological knowledge failed to predict domain-specific analogy scores once age and single word reading was controlled. The finding that vocabulary and phonological knowledge fails to predict children's use of orthographic analogies in beginning reading is inconsistent with previous experimental findings (Bowey & Hansen, 1994; Bowey & Underwood, 1996; Coltheart & Leahy, 1992; Muter, Snowling & Taylor, 1994; Walton, 1995). The more common finding that orthographic analogy use increases with word reading proficiency and vocabulary skill may be an artefact of calculating improvements from pre-test to actual test scores (see also, Wood, 1999, 2000, 2002).

The study revealed important individual differences in children's domain-specific analogy reasoning skills. The analogy profiles were identified based on the speed and accuracy of children's ability to solve orthographic and phonological analogies in early reading. These profiles are particularly important because they explored possible differences in children's reasoning abilities and separated children into different groups according to the efficiency in which they solved orthographic analogies and responded to the phonological prime items. The analogy profiles revealed that one group were fast and accurate in reading words in both the orthographic analogy and phonological prime conditions. Another group showed a similar pattern of results with similar levels of transfer across the analogous and phonological prime conditions but these children were slightly less accurate and slower than children in the previous group. A third group of children were as accurate in reading the orthographic analogy items than children in the intermediate reasoning group, but were less accurate in reading the phonological prime words. A fourth group showed a slight advantage for reading orthographic analogy words than phonological primes but their performance was underscored by slow responses. This was the least accurate and slowest group overall. Thus it seems that the third and fourth groups of children identified were the most attentive to the orthographic similarity between the clue words and target words, and least susceptible to phonological priming effects.

Meaningful relations were found between children's analogy profiles and their word reading and early phonological knowledge. Specifically, we compared profiles of analogy against standardised tests of domain knowledge, to examine the possibility that early reading ability, rhyme awareness and phoneme awareness could explain apparent differences between the analogy groups. An interesting pattern of results emerged

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showing linear trends in domain knowledge across the four groups. As expected, children in the first two groups (high reasoning and intermediate reasoning group) revealed higher levels of single word reading and early phonological skills, than children in the second two groups (orthographic and low reasoning group). Therefore, it appears that increases in overall analogical performance are associated with advances in reading proficiency and early phonological knowledge (*cf.* Goswami, 1993).

We suggest, however, that the identification of different profiles of reasoning by analogy in word reading might have important implications for re-addressing the phonological priming debate. It is possible that the current profiles of individual differences in analogy may underscore qualitative changes in beginning reading. It is apparent that children who fall into the 'higher' analogy groups (high reasoning and intermediate reasoning group) show almost equivalent levels of transfer in reading the orthographic items and phonological primes, which according to Goswami's own strict criterion demonstrates high levels of phonological priming. This indicates that children in these clusters might have been using a phonological approach in their responses to all word pairs, rather than using a more traditional orthographic analogy approach to word reading (see Goswami, 1990). In contrast, children in the 'lowest' groups (the orthographic and low reasoning groups), despite showing poor levels of accuracy and speed overall, were more consistent in the way they read the orthographic words than phonological equivalents. They showed a significantly higher performance when reading orthographic analogy items than when reading the phonological primes, suggesting that the children were using the orthographic information implicit in the clue word to identify unfamiliar target words. Furthermore, this suggests that children in the lowest clusters were in fact more efficient in solving orthographic analogies than the children with the more able profiles. The findings also underscore the possibility that children's use of orthographic analogies in word reading is not associated with increases in reading proficiency and early phonological awareness; rather the ability to make true orthographic analogies based on shared orthographic spelling patterns between words depends on the child's limited knowledge of the reading domain. It also appears that apparent increases in phonological awareness and reading proficiency are in fact associated with the use of phonological priming.

The suggestion that young children make orthographic analogies in the absence of reading skills and phonological knowledge is difficult to explain within existing theoretical models of reading development (*cf.* Goswami, 1993; Goswami & Bryant, 1990). It also challenges previous theoretical claims of a strong positive association between advances in phonological skills and increases in orthographic analogy (Goswami, 1993; Muter, Snowling & Taylor, 1994; Walton, 1995; Wood, 1999, 2000, 2002). The current findings suggest that phonologically skilled children may be overgeneralising a heuristic that they have discovered in which they guess the correct reading of unknown words by means of a purely phonological strategy, based on rime, that appears to 'work' in the context of clue-word-type tasks. In contrast, less phonologically skilled children appear to be relying more heavily on their alphabetic knowledge and on assessing the orthographic consistency between words. Their relative lack of phonological fluency has not yet enabled them to discover the potential of the phonologically based guessing strategy.

What is clear from our analyses is that when employing a more detailed analytical method, which considers individual differences within beginning readers, a more detailed description of children's analogy development does emerge. As the current findings

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suggest, earlier claims by Bowey, Vaughan and Hansen (1998) and Goswami (1988, 1990, 1993) are both correct depending on which sub-group is considered. Using an individual differences approach, we argue that it is possible to clarify the evidence concerning the extent to which phonological priming can account for young children's ability to make orthographic analogies in reading.

A further contribution of the present study was its examination of the relations between general forms of analogy skill and domain-specific analogy in early reading. This concurrent relationship between general reasoning and orthographic analogy use has been reported in at least one previous study (Wood, 1999). Her findings indicated that visual reasoning ability was a significant unique concurrent predictor of orthographic analogy use in beginning reading and raised the possibility that other forms of general analogical reasoning skills may have strong implications for the use of orthographic analogies in reading. In the present analyses, we found that the children's performance on the visual analogical reasoning task could offer some concurrent prediction of children's orthographic analogy scores in beginning reading, although this effect was not strong. This pattern of results did not extend to children's performance on the causal or thematic reasoning tasks. This relationship between visual analogy and orthographic analogies may relate, in particular, to children's ability to recognise visual patterns across both types of task. Children's success in reading by analogy depends on their sensitivity to orthographic similarity and their developing phonological awareness. It is this ability to recognise visual patterns, including the proportion of shared spelling patterns, between words that could explain why the two tasks remain similar (see Goswami, 1998, 1999a). This visual component underlying both the reading analogy task and the visual analogy task may therefore assist in explaining the relationship found here.

If children's sensitivity to visual similarity can best account for the relationship between general reasoning skills and the use of orthographic analogies in beginning reading, then it is hypothesised that this kind of general visual reasoning ability should be able to explain patterns of individual differences in early orthographic and phonological analogies. However, the discriminant function analysis showed that this claim was unsupported. After dividing children into groups according to the efficiency in which they solved orthographic and phonological analogies, group membership was not predicted by their visual analogy skills. Whilst this aspect of general analogical reasoning skill is an important predictor of young children's reading analogy, this measure is not strong enough to predict cluster membership reliably. This finding therefore suggests that orthographic analogy skills in early reading are indicative of domain-specific knowledge rather than a sub-component of a generalised analogical reasoning skill.

In summary, the results support the usefulness of exploring individual differences in children's analogy and demonstrate how early phonological skills play a crucial role in explaining individual differences in children's analogy development (e.g. Goswami, 1993, 1999a; Goswami & Bryant, 1990). More specifically, the findings illustrate the importance of both orthographic and phonological reasoning skills as useful strategies to early word reading. Patterns of individual differences in analogy scores show that whilst some children analogise orthographically, others use phonological priming. The extent to which children use either an orthographic or phonological analogy strategy can be best explained by their knowledge of the reading domain and more importantly their sensitivity to phonological units in early reading. Overall, the findings demonstrate how exploring individual differences in analogy skills is important for understanding children's early reading development.

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## Appendix A

Nonword items used in List A of the Reading Analogy Task (number of times each word read correctly in parentheses).

| Clue words | Orthographic rime analogy | Phonological rhyme analogy | Orthographic beginning analogy | Phonological beginning analogy | Unrelated controls |
|---|---|---|---|---|---|
| List A items | | | | | |
| Sait | Nait (31) | Tayt (29) | Saig (18) | Sayb (14) | Nulk (9) |
| Fean | Hean (29) | Peen (33) | Feap (20) | Feek (22) | Sood (6) |
| Zorn | Gorn (32) | Jawn (24) | Zort (34) | Zawf (19) | Felp (7) |
| Jild | Dild (28) | Syld (22) | Jilm (29) | Jylk (18) | Boct (9) |
| Woon | Poon (38) | Hune (26) | Woot (19) | Wupe (8) | Kasp (9) |
| Solg | Folg (39) | Poulg (29) | Solp (32) | Soulm (18) | Jorm (17) |
| Gurp | Turp (38) | Ferp (34) | Gurm (28) | Gert (26) | Cabe (1) |
| Mout | Sout (31) | Towt (29) | Maug (25) | Mowk (20) | Jisp (8) |

## Appendix B

Nonword items used in List B of the Reading Analogy Task (number of times each word read correctly in parentheses).

| Clue words | Orthographic rime analogy | Phonological rhyme analogy | Orthographic beginning analogy | Phonological beginning analogy | Unrelated controls |
|---|---|---|---|---|---|
| List B items | | | | | |
| Kiym | Niym (29) | Fighm (24) | Kiyd (20) | Kighn (13) | Molp (6) |
| Noad | Hoad (29) | Dode (26) | Noak (22) | Nowf (29) | Gamp (6) |
| Keyl | Leyl (34) | Yail (29) | Keyn (22) | Kaif (18) | Wask (3) |
| Gort | Jort (33) | Lawt (37) | Gorf (32) | Gawb (15) | Tesp (5) |
| Kurp | Nurp (39) | Herp (43) | Kurn (31) | Kerf (32) | Helt (6) |
| Tolt | Nolt (36) | Doult (30) | Tolp (34) | Towlk (20) | Shim (13) |
| Noil | Doil (36) | Poyle (29) | Noim (15) | Noyk (16) | Keeb (9) |
| Goot | Doot (27) | Luwt (31) | Goop (33) | Gouk (20) | Faif (5) |

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